Intelligent Ground Vehicle Competition 2026

AutoNav & Self-Drive

Cornell Electric Vehicles

Cornell University

Design Report

[Vehicle Name]

[Vehicle Photo / Sketch]

[Faculty Advisor's Signed Statement of Integrity]

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Contents

1. System Requirements and Design

Things to Write

- Describe the system engineering process you followed to identify important System and Subsystem requirements.
- List at least 2 requirements for: (a) the mechanical design of your vehicle, (b) safety, (c) the electrical/electronic components
- For each challenge you are targeting (AutoNav and SelfDrive), list at least two requirements that guided your design for each of: (a) perception, (b) driving logic, and (c) key performance indicators.
- For each requirement you listed (12 for one course, 18 if both) explain what in these rules elsewhere that drives this requirement, describe how you will quantifiably measure the requirement, and set a target value.

2. Mechanical Design

Things to Write

- Describe the mechanical design of your robot, including CAD or other drawings of your vehicle
- Describe significant decisions on the frame structure, housing, and/or overall design
- Describe significant components of your vehicle's drive system, electrical power system, and the control system that enables computer-controlled motion. Components acquired ready-made must be identified, but their internal components need not be described in detail.
- Describe the significant mechanical components, including the frame, structure, suspension, housing, and weather proofing. Clearly identify the aspects of your mechanical design that are driven by each mechanical system requirement you listed.
- For each mechanical requirement you listed, compare the target value with the most recently measured actual value, and discuss what this tells you about your vehicle

3. Safety

Things to Write

- Describe the robot's safety aspects while being transported, parked, and charging.
- Describe the robot's safety aspects while operating on the course(s), including the mechanical and wireless ESTOP systems.
- Clearly identify the aspects of your safety efforts that are driven by each safety requirement you listed.
- For each safety requirement you listed, compare the target value with the most recently measured actual value, and discuss what this tells you about your vehicle

4. Electrical/Electronic Design

Things to Write

- Describe significant power, electrical, and electronic components in your vehicle, including electronics, computers, sensors, and motor controllers
- Describe significant decisions made in designing/selecting the electrical and electronic components
- Describe the power distribution system's capacity, max. run time, recharge time, and safety
- Clearly identify the aspects of your electrical and electronic components that are driven by each electrical/electronic components system requirement you listed.
- For each electrical/electronic components system requirement you listed, compare the target value with the most recently measured actual value, and discuss what this tells you about your vehicle

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5. Perception

Things to Write

- If you are targeting both challenges, you must include separate discussions of how your perception system matches each course.
- Describe processing of the sensor data to identify course features and items around the vehicle, including obstacles and lane markings, and the use of machine vision to perceive the environment.
- If targeting AutoNav, describe how the system identifies lanes and obstacles.
- If targeting SelfDrive, describe how the vehicle senses pedestrians, stop signs, and road obstacles.
- Describe the internal representation of the course built from the perception data which is used by the driving logic. This might be a sophisticated internal representation of the environment or a simple list of features. Include a discussion of how it is created and updated as the robot moves.
- Clearly identify how your design of the sensor data processing system was driven by the perception requirements you listed.
- For each perception requirement you listed, compare the target value with the most recentlymeasured actual value, and discuss what this tells you about your vehicle
- Developed a custom C++ Iterative Closest Point (ICP) node to process LiDAR point clouds for robust spatial mapping.
- Implemented a 3D-to-2D vertical slicing filter (using ROS 2 pointcloud iterators) to compress 3D LiDAR data, isolating course obstacles at specific Z-heights while filtering out ground and ceiling noise to prevent false obstacle detection.

6. Driving Logic

Things to Write

- If you are targeting both challenges, you must include separate discussions of how your driving logic matches each course.
- Design of the lane following and obstacle avoidance systems must be specifically described, along with how the vehicle navigates through its environment. Describe how the system uses GPS for waypoint navigation and localization.
- If targeting AutoNav, describe how your robot uses the perception system's output and other information to follow lanes, detect and avoid obstacles, climb the ramp, and navigate to the GPS waypoints. Include how vehicle motion is generated and monitored to move the vehicle through the course, including how it deals with complex obstacles including switchbacks and center islands dead ends, traps, and potholes.

- If targeting SelfDrive, describe how the vehicle uses the perception system's output and otherinformation to follow the rules of the road while navigating the course that includes pedestrians,stop signs, road obstacles, and other items.
- Clearly identify how your design was driven by the driving logic requirements you listed.
- For each driving logic requirement you listed, compare the target value with the most recentlymeasured actual value, and discuss what this tells you about your vehicle

Extended Kalman Filter

I think that this goes here, if not we can move it to the "Driving Logic" section

Main things to higlight here:

- JAX implementation of EKF and *high level* of how dynamics were formulated
- System assumptions and why they were made
- Performance evaluation and observability of the system as a whole
- Link to repository

7. Key Performance Indicators

Things to Write

- If you are targeting both challenges, you must include separate discussions for each course.
- Describe how you will measure each of the key performance indicators you selected to check that your robot is competitive on the target course(s), what value/result you are targeting, and how you selected that value/result for the KPI.
- For each KPI requirement you listed, compare the target value with the most recently measured actual value, and discuss what this tells you about your vehicle

8. Analysis of Complete Vehicle

Things to Write

- Describe lessons learned during construction and system integration
- Identify key mechanical and electrical components that have failed during testing and discuss what changes you made to prevent repeat failures. Did your changes prevent repeat failures?
- Describe your software testing, bug tracking, and version control processes
- Describe your simulation-based testing process (e.g., Gazebo, etc.)
- Describe physical testing to date (mechanical, electronic, indoors, real world, etc.) and identify key differences in actual performance compared to predictions from simulation.

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9. Cyber Security Analysis

Things to Write

- Modern robotic and human driven vehicles are network-connected, contain large amounts of software from multiple vendors, and are an inviting target for cyber-attacks.

- List 3 cyber vulnerabilities present in your vehicle and discuss what steps should be taken to harden/remove them before it enters series production and is widely deployed.